(T) Roll No. .....

PAPER ID—21133

Bachelor of Technology (Computer Science and Engineering (Artificial Intelligence and Machine Learning)) Examination, 2025

(Fourth Semester)

Design and Analysis of Algorithms

Time : 3 Hours *Maximum Marks : 70*

Before answering the question-paper candidates should ensure that they have been supplied to correct and complete question-paper. No complaint, in this regard, will be entertained after the examination.

Note : Attempt *Five* questions in all, selecting *one* question from each Unit. Q. No. **1** is compulsory. All questions carry equal marks.

1. (a) What is space complexity of an algorithm ?

(b) What are disjoint sets ?

(c) Explain single source shortest path.

(d) Compare small Oh and small Omega notations.

(e) Briefly explain graph coloring.

(f) Write note on simplified NP hard problem.

(g) What are limitations of back tracking ?

7×2=14

Unit I

2. Write and explain Strassern's matrix multiplication algorithm along with its complexity. 14

3. What is Selection Sort ? Write the Selection Sort Algorithm. Sort the following array by using selection sort : 14

24, 84, 37, 12, 73, 67, 54, 98, 44

Unit II

4. Suggest an approximation algorithm for travelling salesperson problems using minimum spanning tree algorithm. Assume that the cost function satisfies that triangle inequality. 14

5. What is Dynamic Programming ? Also Explain binary search tree by using a suitable example. 14

Unit III

6. Explain the following back tracking techniques : 14

(a) Graph coloring

(b) Hamiltonian cycle.

7. List and explain the steps to solve 0/1 knapsack problem with branch and bound technique. **14**

Unit IV

8. Discuss the NP Hard code generation problem. **14**

9. Compare NP Hard and NP Complete problem. **14**